Detecting Recyclable Waste on a Conveyor Belt with YOLOv8: A Comparison of Data Augmentation Policies

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***Abstract*—Collection centers receive mixed waste whose manual sorting is slow and costly. This work proposes a computer vision system that, installed over a conveyor belt, detects and classifies waste into six categories (biodegradable, cardboard, glass, metal, paper and plastic) in order to prioritize the extraction of the materials with the highest recycling value. We trained and compared four configurations – two YOLOv8 variants (n and s) under two data augmentation policies – on a dataset of 10,464 images (Garbage Classification 3, Roboflow). The proposed augmentation policy, designed from a quantitative analysis of the dataset's shortcomings (orientation, lighting and sharpness), scored 1.2 points (YOLOv8s) and 0.7 points (YOLOv8n) of mAP@0.5:0.95 above the default Ultralytics configuration in single training runs, with YOLOv8s + proposed reaching 0.650 mAP@0.5 and 55 FPS on a T4 GPU.**

***Index Terms*—object detection, YOLOv8, waste classification, data augmentation, computer vision, recycling.**

I. Introduction

Busy cities generate large amounts of mixed waste. Sorting it manually, material by material, is slow and costly, and much of the recyclable material loses commercial value because it is not identified and extracted in time during processing. Classifying directly on the street during collection is impractical because of lighting and motion conditions; it is more viable to automate classification once the waste reaches a collection center or processing plant, where it can pass along a conveyor belt under controlled camera conditions.

The goal of this work is to design and evaluate a deep-learning-based object detection system that, over a conveyor belt, identifies the category of each item in real time and allows the extraction of the highest-value materials (for example aluminum, PET or cardboard) to be prioritized, instead of requiring exhaustive separation of 100% of the flow. The scope of this work is the computer vision problem — detecting and classifying the different types of waste — leaving the physical implementation of the separation mechanism (a robotic arm or compressed air) as a future extension.

As a real-world precedent, Koskinopoulou et al. [1] describe a computer vision categorization system for the industrial robotic separation of recyclable waste, deployed and evaluated in a real waste processing plant under demanding industrial conditions, using a low-cost deep-learning-based vision module. In industry, the trade press reports a deployment of a Recycleye AI picking robot at the re3 materials recovery facility (MRF) in Reading, United Kingdom, with 99% purity of the separated material versus 96% on the human reference sorting line and up to 55 picks per minute [2]. Both precedents support the viability of a computer vision approach to this task, although the industry figures are not independently audited, unlike the metrics reported in this work.

II. Methodology

A. Dataset and Processing

We used the public Garbage Classification 3 dataset (v2, YOLOv8 format) from Roboflow [3], with 10,464 images annotated in six classes: biodegradable, cardboard, glass, metal, paper and plastic. Before training, the raw export was audited to discard corrupt images and to fix or remove defective annotations — a necessary step because Roboflow exports can contain labeling errors that silently harm training. The dataset was re-split into our own 70/20/10 (train/val/test) partition instead of keeping Roboflow's, which was class-imbalanced and would have invalidated the per-class evaluation. The test set was kept held out and unaugmented until the final evaluation.

B. Augmentation Policy

The augmentation techniques were chosen from a quantitative analysis of the processed dataset — not from generic assumptions — measuring brightness, sharpness, color and class balance. The analysis showed that the dataset lacks dark or blurry images, conditions expected in real conveyor belt video, and that the observed class imbalance is due to the number of objects per photo rather than the number of photos per class. From these findings we defined two configurations (*arms*) to compare: `ultralytics_defaults`, the default Ultralytics augmentation [4] (mosaic, HSV jitter, horizontal flip, translation and scale), and `proposed`, which keeps those defaults and adds vertical flip, darkening (brightness reduction of up to 40% with probability 0.3), motion blur and Gaussian blur, to cover the measured gaps in orientation, lighting and sharpness, implemented with Albumentations [5]. When custom Albumentations transforms are supplied, Ultralytics replaces its own built-in set (Blur, MedianBlur, ToGray and CLAHE, each with probability 0.01) instead of adding to

it, so `proposed` also drops those four transforms, which `ultralytics_defaults` keeps. Augmentation is applied *online* during training from a single shared configuration file, so all runs use exactly the same policy and the test set is never augmented by accident.

C. Training and Evaluation

We fine-tuned two pre-trained YOLOv8 variants [4] — YOLOv8n and YOLOv8s — under each of the two augmentation policies, yielding four trained configurations. Each model was trained for 50 epochs with batch size 16 and a fixed seed (42) on a T4 GPU (Google Colab). Evaluation was performed on the held-out test set, reporting mAP@0.5, mAP@0.5:0.95, F1 and end-to-end inference latency (200 test images, processed one at a time), measured on both a T4 GPU and a laptop CPU (Intel i5-1135G7). F1 is the harmonic mean of the mean precision and mean recall reported by Ultralytics.

III. Experimental Results and Analysis

Table I summarizes the metrics of the four configurations on the test set. The `proposed` policy scored higher than `ultralytics_defaults` on both architectures: +1.2 points of mAP@0.5:0.95 for YOLOv8s and +0.7 points for YOLOv8n, with the same ordering on the validation set (+1.4 and +1.1 points). It was higher in five of the six classes on both models, with a single minor exception in CARDBOARD (-0.008). With one training run per configuration, this is a consistent direction rather than a statistically tested difference. Nor can the gain be attributed to a specific transform: the two recipes differ in vertical flip, darkening, blur and the four built-in Ultralytics transforms that `proposed` drops.

TABLE I
Metrics by model and recipe (test set)

Model	Recipe	mAP@0.5	mAP@.5:.95	F1
YOLOv8s	proposed	0.650	0.467	0.650
YOLOv8s	ultralytics_defaults	0.645	0.455	0.646
YOLOv8n	proposed	0.629	0.443	0.625
YOLOv8n	ultralytics_defaults	0.622	0.435	0.617

YOLOv8s outperformed YOLOv8n in all six classes under both recipes, at the cost of higher latency: on the T4 GPU, YOLOv8s + proposed processed each image in 18.1 ms (55 FPS) versus 10.2 ms (98 FPS) for YOLOv8n + proposed. On CPU (Intel i5-1135G7, `proposed` weights), YOLOv8n reached 68.9 ms (14.5 FPS) versus 177.3 ms (5.6 FPS) for YOLOv8s — a 2.6× difference that matters for deployment without a dedicated GPU.

The recommended model is YOLOv8s with the `proposed` policy, as it has the best mAP@0.5:0.95 and F1 of the set and runs in real time on GPU (55 FPS). For deployment without a GPU we recommend YOLOv8n + `proposed` as an alternative, with a 2.5-point drop in mAP@0.5:0.95 in exchange for being 2.6× faster on CPU. The worst-performing classes on both models are BIODEGRADABLE, CARDBOARD and PAPER (mAP@0.5:0.95 between 0.36 and 0.43 for YOLOv8s); the next subsection examines why.

A. Error Analysis on the Test Set

Fig. 1 shows the confusion matrix of the recommended model on the test set at confidence 0.25, the threshold used at inference and in the demonstration below. Confusion between materials is rare: the largest off-diagonal entry is METAL predicted as PLASTIC (5.5% of METAL objects), and no other pair exceeds 3.3%. In particular, BIODEGRADABLE, CARDBOARD and PAPER are almost never mistaken for one another (at most 2.8%, PAPER predicted as CARDBOARD). The dominant error is missed objects: 51% of CARDBOARD, 42% of BIODEGRADABLE and 40% of PAPER objects are not detected, against 22–28% for GLASS, METAL and PLASTIC. This matches the gap between precision (0.743) and recall (0.577) in Table I. BIODEGRADABLE also accounts for 55% of the false detections on background; it is by far the most densely annotated class, with 4,504 of the 7,452 test boxes (60%). The weak classes are therefore limited by detection, not by confusion between materials.

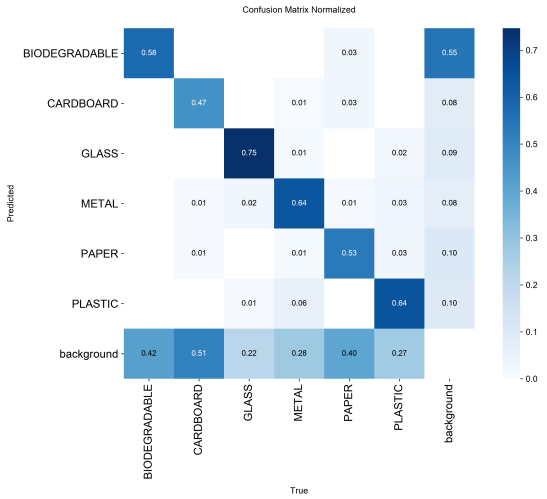


Fig. 1. Normalized confusion matrix of YOLOv8s + `proposed` on the test set at confidence 0.25 (columns: true class). The bottom row is the fraction of objects that were not detected; the right column distributes the false detections on background.

B. Demonstration on a Simulated Belt

To illustrate how the system works without a physical belt, we built a simulation: 11 images from the test set (randomly chosen with a fixed seed), each resized to a 420 px square, slide from left to right over a 1280×540 px belt background at 30 FPS for 20 s (600 frames), as if passing in front of a fixed camera. The recommended model (YOLOv8s + `proposed`, confidence threshold 0.25) runs frame by frame, as an online system would. Fig. 2 shows six representative frames.

Per-frame latency (preprocessing, inference and NMS), measured after 10 untimed warm-up frames on an integrated GPU (MPS backend), was 6.35 ms on average (157 FPS), with

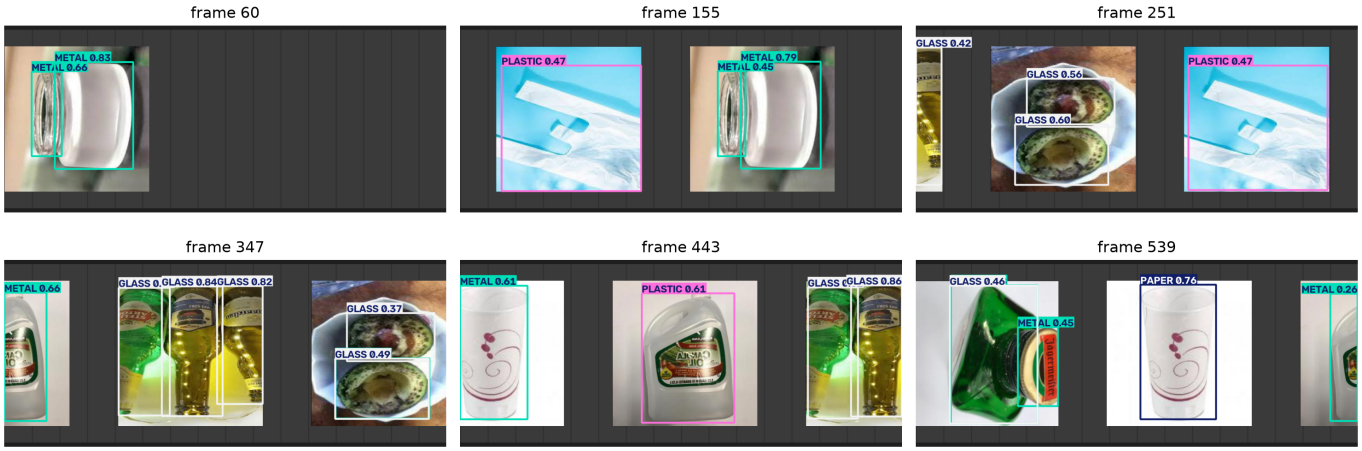


Fig. 2. Six frames of the simulated belt demonstration, with the detections of the YOLOv8s + `proposed` model. Correct detections (glass bottles, paper cup) and class confusions, particularly METAL, can be seen.

a median of 6.45 ms and a 95th percentile of 7.09 ms, well above the 30 FPS of the simulated video.

TABLE II
DETECTIONS PER CLASS IN THE DEMONSTRATION (PER-FRAME COUNT)

Class	Detections	Mean confidence
GLASS	1241	0.616
METAL	746	0.617
PLASTIC	482	0.487
PAPER	188	0.686
BIODEGRADABLE	53	0.554
CARDBOARD	5	0.396

Table II counts detections *per frame*: the same object appears in many consecutive frames, so the table reflects the images that traveled along the belt and not the number of distinct objects (which would require tracking; see Conclusions). In the inspected frames we observed two behaviors. On one hand, objects with a clear appearance are classified with high confidence, for example the upright glass bottles (0.82 to 0.86). On the other hand, there are class confusions: the same white plastic jug was labeled PLASTIC (0.61) in one frame and METAL (0.66) in another; a paper cup was labeled PAPER (0.76) and, when entering from the edge of the belt partially cropped, METAL (0.61); a glass jar was labeled METAL; and avocado scraps in a bowl were labeled GLASS instead of BIODEGRADABLE. The class becomes unstable when the object appears cut off at the edge of the frame. CARDBOARD was almost never detected (5 detections, mean confidence 0.396), consistent with its low test performance. These observations are qualitative and based on 11 distinct images, so they do not replace the quantitative evaluation of Table I.

Limitations: each model and recipe was trained only once, so the reported differences were not subjected to a statistical significance test. The two recipes differ in several transforms at once, so the comparison does not isolate the effect of any single one, and they were trained with consecutive Ultralytics releases (8.4.173 for `ultralytics_defaults` and 8.4.172 for `proposed`). Of the four recipes defined in the augmentation policy, only two (`ultralytics_defaults` and `proposed`) were actually trained; `no_augmentation` and `proposed_low_saturation` remain as future work. Moreover, the dataset used, although realistic, does not fully replicate the level of visual clutter (overlapping objects, partial occlusions) that the specialized literature documents in real conveyor belt video [6]; validating the recommended model against our own belt images, closer to that scenario, is a natural step before a production deployment.

IV. CONCLUSIONS

1) An augmentation policy designed from quantitatively measured dataset gaps (orientation, lighting, sharpness) scored higher mAP@0.5:0.95 than the Ultralytics defaults on both architectures and on both validation and test, without collecting more data. With one run per configuration and recipes that differ in several transforms, confirming the effect and attributing it to specific transforms requires repeated runs and an ablation.
2) YOLOv8s offers better accuracy than YOLOv8n in all six classes under both recipes, but at a relevant latency cost on CPU (2.6× slower than YOLOv8n), so the choice between them depends directly on the hardware available at the collection center.
3) Re-splitting the dataset instead of using Roboflow's original partition was necessary to obtain a reliable per-class evaluation, given the imbalance detected in the original partition.
4) The BIODEGRADABLE, CARDBOARD and PAPER classes concentrate the lowest performance of the system, and their main error on the test set is missed objects (40–51% undetected at confidence 0.25), not confusion with other materials (at most 2.8%). Improving recall on these classes, for example in densely annotated

BIODEGRADABLE images, is the main bottleneck to address in a future iteration.

5) The simulated belt demonstration ran at 157 FPS on an integrated GPU (MPS), well above the 30 FPS of the video, but showed confusions, mainly METAL versus PLASTIC, PAPER and GLASS, and unstable classes when the object is cut off at the edge of the frame. This reinforces that the next step is validating with our own belt images and with partially visible objects.
6) As future work, the architecture can be extended with object tracking to avoid counting the same item twice in video, and by directly evaluating the physical implementation of the separation mechanism, which was outside the scope of this work.

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